Higher National Diploma in Engineering (Mechanical/Electrical)
Forth Year First semester Examination- 2022
ME3227 – INDUSTRIAL ENGINEERING MANAGEMENT

Instructions for Candidates:
Answer any 5 questions.

No. of questions : 06
No. of pages : 03
Time : 03 hours

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01)

a) Why identification of problem is important when OR is used? (02 marks)

b) A manufacturing facility produces two types of toys, A & B. The production line of the facility has 2 machines for row material processing each having 12 hr of operating time per day, 5 workers in assembly department and 2 workers in finishing and packing department each having 6 hr per day duty schedule. The required time duration for 10 nos. of each product to process before go to the market is as follows.

Type of toy	Production time (hours per box)			Profit per 10 items (Rs.)
	Row material processing	Assembly department	Packaging and Shipping department	
A	2	3	1	10/=
B	4	2	1	15/=
Total	24	30	12	

Assuming that the company wants to maximize profit,

i) Formulate Linear Programming model. (06 marks)

ii) By using the graphical method find the number of each type of toy, that company should manufacture per day in order to maximize the profit. (10 marks)

iii) What is the maximum profit that the company can achieve? (02 marks)

02)

A) a) The price of a particular good was Rs.4000/= and Rs.2000/= when the quantity supplied was 80 & 60 respectively. Calculate the price elasticity of supply. (05 marks)

b) Discuss the disadvantages of monopoly market structure giving suitable examples from Sri Lankan market. (05 marks)

B) a) Name any 4 activities of project life cycle. (04 marks)

b) Describe any 2 of above activities. (06 marks)

03)

a) A printing machine was purchased at a price of Rs.200,000/= and the service charge as per the agreement was Rs.20,000/= at start of every year for 5 years of effective life time. The annual income of the machine was Rs.80,000/= per year and the salvage value was Rs.30,000/= Draw a net cash flow diagram for the life time of the machine. (04 marks)

b) The production department of a factory, is planning to purchase a new processing machine and they receive 2 options to implement their purchase.

Option 01 - The full payment of Rs.400,000/= for the machine will be done initially. The service will be provided for 5 yrs life time, under the agreement for first 4 years at a service charge of Rs.50,000/= and 5th yr free of charge.
The estimated annual income for the 1st year will be Rs.180,000/= and for next consecutive 4 years it will be increased by 10%, compared to the previous year.

Option 02 - The payment for the machine will be done in 5 installments of Rs.150,000/= including service charges. According to the agreement Rs.150,000/= will be payed initially and remaining 4 annual installments will be paid during the first 4 yrs. The annual income for 1st year is estimated as Rs.175,000/= and for next consecutive 4 years it will be increased by 10%, compared to the previous year. The salvage value at the end of 5 year period will be Rs.50,000/=.

i) Calculate the yearly Net cash flows of the two alternatives separately. (06 marks)

ii) Calculate the NPV for purchasing of both machines at a rate of 10% interest. (08 marks)

iii) Which alternative will be the best option for the company and why do you recommend it? (02 marks)

04)

a) What are the limitations and advantages of functional organization structure? (04 marks)

b) Describe the use of having proper organizational structure for an organization, clearly mentioning the advantages of it. (06 marks)

c) Name any 4 types of key Human Resource activities. (04 marks)

d) Why it is required to have very good leadership skills as a manager to enhance the motivation of employees of the company? Describe with relevant leadership characteristics. (06 marks)

05)

a) What is the meaning of quality? (02 marks)

b) Distinguish the difference between quality of design and quality of conformance (quality of manufacture). (06 marks)

c) Name any 04 quality control tools. (04 marks)

d) Describe any 02 of above quality control tools. (08 marks)

06)

a) Give 04 advantages of process layout in facility layout design. (04 marks)

b) The activities of a production line on assembling a particular product is as given in the table below. The total production time available per day is 18 hours and the output is 90 units per day.

Task	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	A	A	A,B	C	D	C,D,F	G	H,E	F	I,J
Task time (min)	6	6	4	8	10	4	3	2	5	4	8

i) Draw the network diagram. (06 marks)
ii) Compute the cycle time per workstation in minutes. (02 marks)
iii) Compute the minimum number of workstations. (02 marks)
iv) How would you combine the tasks into workstations? (04 marks)
v) What is the efficiency of line balancing? (02 marks)